

Inverse Correlation Similarity: Algorithm D

Precomputed Basis Construction, Theory, and Applications

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1 Introduction

Algorithm C solved the inverse correlation problem using stochastic orthogonalization and centered geometric decomposition.

In many practical applications, however, the anchor vector remains fixed while the target correlation changes repeatedly across large numbers of queries. Examples include:

- Monte Carlo covariance simulation.
- Synthetic financial return generation.
- High-frequency stress testing.
- Streaming synthetic dataset construction.
- Correlation-controlled adversarial perturbation.

In such settings, repeatedly recomputing centered orthogonal projections becomes computationally inefficient.

This paper presents Algorithm D, a precomputed-basis algorithm for solving the inverse correlation problem with amortized query acceleration.

The central idea is to precompute an orthonormal basis for the centered hyperplane

$$H = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0 \right\},$$

whose first basis vector equals the normalized centered anchor vector.

After the setup phase, each query requires only:

- one square root,
- two scalar-vector multiplications,
- one vector addition.

No random number generation, orthogonalization, normalization, or dot products are required at query time.

The paper ends with “The End”.

2 Background and Notation

2.1 Pearson Correlation

Definition 2.1 (Pearson Correlation). *Let*

$$u, v \in \mathbb{R}^n$$

be nonconstant vectors.

Define the sample means

$$\bar{u} = \frac{1}{n} \sum_{i=1}^n u_i, \quad \bar{v} = \frac{1}{n} \sum_{i=1}^n v_i.$$

The Pearson correlation coefficient is

$$\rho(u, v) = \frac{\sum_{i=1}^n (u_i - \bar{u})(v_i - \bar{v})}{\sqrt{\sum_{i=1}^n (u_i - \bar{u})^2} \sqrt{\sum_{i=1}^n (v_i - \bar{v})^2}}.$$

2.2 Centering Matrix

Define the centering matrix

$$P = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top,$$

where

$$\mathbf{1} = (1, \dots, 1)^\top.$$

Theorem 2.2. *The matrix*

$$P$$

is symmetric and idempotent:

$$P^\top = P, \quad P^2 = P.$$

Proof. Symmetry follows immediately:

$$P^\top = \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right)^\top = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top = P.$$

Now compute

$$P^2 = \left(I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top \right)^2.$$

Expanding:

$$P^2 = I - \frac{2}{n} \mathbf{1} \mathbf{1}^\top + \frac{1}{n^2} \mathbf{1} (\mathbf{1}^\top \mathbf{1}) \mathbf{1}^\top.$$

Since

$$\mathbf{1}^\top \mathbf{1} = n,$$

we obtain

$$P^2 = I - \frac{1}{n} \mathbf{1} \mathbf{1}^\top = P.$$

□

2.3 Centered Hyperplane

The image of

$$P$$

is the mean-zero hyperplane

$$H = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0 \right\}.$$

This is an

$$(n-1)$$

-dimensional subspace of

$$\mathbb{R}^n.$$

2.4 Correlation as Cosine Similarity

Define centered vectors

$$u_c = Pu, \quad v_c = Pv.$$

Then

$$\rho(u, v) = \frac{u_c^\top v_c}{\|u_c\| \|v_c\|}.$$

Thus Pearson correlation equals cosine similarity inside the centered hyperplane.

3 The Inverse Correlation Problem

Definition 3.1 (Inverse Correlation Problem). *Given*

$$u \in \mathbb{R}^n$$

and target correlation

$$c \in [-1, 1],$$

construct

$$v \in \mathbb{R}^n$$

such that

$$\rho(u, v) = c.$$

4 Algorithm D

Algorithm D consists of two phases:

- Setup phase.
- Query phase.

4.1 Setup Phase

Compute the centered anchor vector

$$u_c = Pu.$$

Normalize:

$$q_1 = \frac{u_c}{\|u_c\|}.$$

Construct an orthonormal basis

$$Q = [q_1 \quad q_2 \quad \cdots \quad q_{n-1}]$$

for the centered hyperplane

$$H.$$

Thus

$$q_i^\top q_j = \delta_{ij}.$$

The basis may be constructed using:

- Householder reflections.
- QR decomposition.
- Modified Gram–Schmidt orthogonalization.

4.2 Query Phase

Given target correlation

$$c \in [-1, 1],$$

choose basis vector

$$q_j, \quad j \in \{2, \dots, n-1\}.$$

Define

$$v_c = cq_1 + \sqrt{1-c^2} q_j.$$

Finally define

$$v = v_c + \mu \mathbf{1},$$

where

$$\mu \in \mathbb{R}$$

is arbitrary.

4.3 Batch Query Phase

For target correlations

$$c_1, \dots, c_K,$$

construct vectors

$$v_1, \dots, v_K$$

using

$$v_k = c_k q_1 + \sqrt{1-c_k^2} q_{j_k},$$

where

$$j_k \in \{2, \dots, n-1\}.$$

5 Theoretical Analysis

5.1 Correctness

Theorem 5.1 (Correctness of Algorithm D). *Algorithm D produces a vector*

$$v$$

satisfying

$$\rho(u, v) = c.$$

Proof. By orthonormality,

$$q_1^\top q_j = 0, \quad \|q_j\| = 1.$$

Hence

$$q_1^\top v_c = q_1^\top (cq_1 + \sqrt{1-c^2} q_j) = c.$$

Also,

$$\|v_c\|^2 = c^2 + (1-c^2) = 1.$$

Therefore

$$\frac{u_c^\top v_c}{\|u_c\| \|v_c\|} = c.$$

Since correlation equals cosine similarity of centered vectors,

$$\rho(u, v) = c.$$

□

5.2 Complexity

Theorem 5.2 (Complexity of Algorithm D). *Algorithm D has the following computational complexity:*

<i>Phase</i>	<i>Time Complexity</i>	<i>Space Complexity</i>
<i>Setup</i>	$O(n^2)$	$O(n^2)$
<i>Query</i>	$O(n)$	$O(1)$
<i>Batch Query</i>	$O(Kn)$	$O(1)$

Proof. Constructing the orthonormal basis requires orthogonalization of

$$n - 1$$

vectors in dimension

$$n.$$

This requires

$$O(n^2)$$

time and storage.

Each query requires:

- one square root,
- two scalar-vector multiplications,
- one vector addition.

Each vector operation requires linear time in dimension

$$n.$$

Hence query complexity is

$$O(n).$$

□

5.3 Amortized Query Cost

Corollary 5.3. *For*

$$M$$

queries against the same anchor vector, the amortized complexity per query equals

$$O(n) + O\left(\frac{n^2}{M}\right).$$

As

$$M \rightarrow \infty,$$

the setup cost becomes negligible.

5.4 Space-Time Trade-Off

Algorithm D admits a space-time trade-off analogous to Algorithm B.

- Explicitly storing the basis requires $O(n^2)$ memory but minimizes query overhead.
- Implicit storage reduces memory usage but increases reconstruction cost during queries.

6 Geometric Interpretation

Algorithm D operates entirely inside the centered hyperplane

$$H.$$

The valid outputs lie on the centered angular shell

$$S^{n-3} \subset H.$$

The centered output vector decomposes into:

- a parallel component

$$cq_1,$$

- an orthogonal component

$$\sqrt{1 - c^2} q_j.$$

These components combine to produce the desired correlation exactly.

7 Comparison with Algorithm C

Feature	Algorithm C	Algorithm D
Orthogonalization	Per Query	Precomputed
Randomness	Yes	No
Setup Cost	Minimal	$O(n^2)$
Query Cost	$O(n)$	$O(n)$
Random Number Generation	Required	Not Required
Dot Products at Query Time	Required	Not Required
Normalization at Query Time	Required	Not Required
Best Use Case	Few Queries	Many Queries

Algorithm D exchanges setup cost and memory usage for improved repeated-query efficiency.

8 Numerical Properties

Algorithm D achieves machine-precision correlation accuracy because:

- the basis vectors are orthonormal,
- the output vector norm is analytically controlled,
- no iterative optimization is required.

Consequently, the algorithm avoids numerical drift associated with repeated stochastic orthogonalization.

9 Applications

9.1 Finance

Applications include:

- synthetic return generation,
- covariance stress testing,
- factor model simulation,
- scenario generation.

9.2 Machine Learning

Applications include:

- synthetic dataset generation,
- adversarial perturbation,
- covariance benchmarking,
- feature-space exploration.

9.3 Econometrics

Applications include:

- multicollinearity analysis,
- regression stress testing,
- covariance simulation,
- estimator sensitivity analysis.

10 Extensions

10.1 Multiple Anchor Vectors

Given anchor vectors

$$u_1, \dots, u_L,$$

store separate centered orthonormal bases

$$Q^{(1)}, \dots, Q^{(L)}.$$

10.2 Correlation Matrix Synthesis

Given target correlation matrix

$$R \in \mathbb{R}^{m \times m},$$

construct dataset

$$X \in \mathbb{R}^{n \times m}$$

such that

$$\text{Corr}(X) = R.$$

This connects Algorithm D to:

- covariance simulation,
- Gaussian copulas,
- Cholesky factorization,
- random matrix theory.

10.3 GPU Parallelization

The batch query phase is trivially parallelizable because each query requires only independent vector assembly operations.

11 Conclusion

We presented Algorithm D, a precomputed-basis algorithm for solving the inverse correlation problem efficiently across repeated queries.

The key insight is that Pearson correlation equals cosine similarity inside the centered hyperplane. By precomputing an orthonormal basis for that hyperplane, Algorithm D eliminates all query-time orthogonalization, normalization, and random number generation.

Applications span:

- finance,
- machine learning,
- econometrics,
- statistical simulation,
- synthetic data generation.

Algorithm D suggests the possibility of a broader theory of inverse statistical geometry in which statistical dependence structures become explicitly constructible objects.

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Glossary

Centered Hyperplane The subspace

$$H = \left\{ x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 0 \right\}.$$

Centering Matrix The projection operator

$$P = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top.$$

Correlation Shell The set of vectors having prescribed correlation relative to a fixed anchor vector.

Inverse Correlation Problem The problem of constructing vectors with prescribed Pearson correlation.

Orthogonal Basis A set of mutually orthogonal unit vectors spanning a subspace.

Target Correlation The prescribed value

$$c = \rho(u, v).$$

The End